# Practice workbook · Volume of revolution about the x-axis

Squaring y and finding pi times the integral of y^2 dx between limits, including solving for an unknown limit.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-22

# 1. Question 6 · part b · 6 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part b only.

**6** (a) Using a formula from page 2, show that $\cos 2A = 1 - 2\sin^2 A$

**(2)**

The finite region $R$ is bounded by the curve with equation $y = 3 + 2\sin x$, the $x$-axis, the $y$-axis and the line with equation $x = \frac{\pi}{4}$

The region $R$ is rotated through 360° about the $x$-axis.

(b) Use calculus to find the volume of the solid generated.
Give your answer to the nearest integer.

**(6)**

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P 7 1 6 6 6 A 0 1 6 3 6

## 1. Question 6 · part b · 6 marks

2022 JUN · Paper 02

Worked-example retry · Attempt part b only.

**Question 6 continued**

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**(Total for Question 6 is 8 marks)**

## 1. Question 6 · part b · 6 marks · Working and answer

Working and answer

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## 2. Question 11 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

**11** Given that

$$y = \frac{e^{4x}}{32}(8x^2 - 4x + 1)$$

(a) show that $\frac{dy}{dx} = x^2e^{4x}$

**(5)**

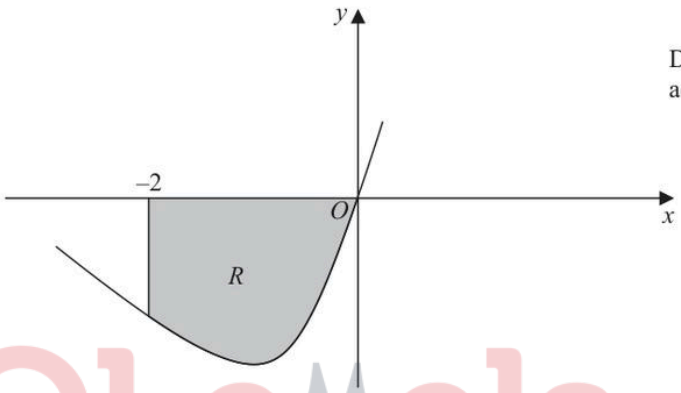



Diagram **NOT** accurately drawn

**Figure 2**

Figure 2 shows part of the curve $C$ with equation $y = 3xe^{2x}$

The finite region $R$ bounded by $C$, the straight line with equation $x = -2$ and the $x$-axis, shown shaded in Figure 2, is rotated though 360° about the $x$-axis.

(b) Using part (a), find the volume, to 2 significant figures, of the solid formed.

**(4)**

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## 2. Question 11 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

**Question 11 continued**

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## 2. Question 11 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

**Question 11 continued**

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**(Total for Question 11 is 9 marks)**

## 2. Question 11 · part b · 4 marks · Working and answer

Working and answer

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## 3. Question 11 · part whole · 7 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

**11**

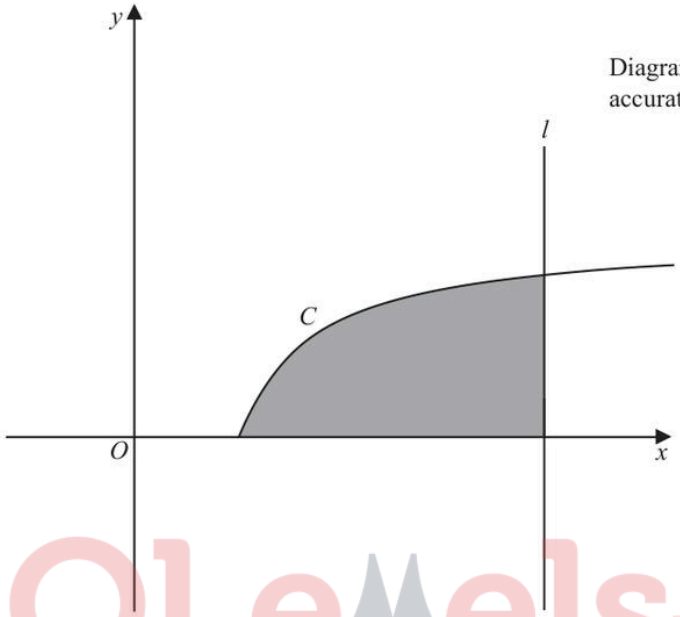



**Figure 5**

Figure 5 shows part of the curve $C$ with equation $y = \sqrt{4x - 8}$ and the line $l$ with equation $x = b$ where $b > 0$

The finite region bounded by the curve $C$, the $x$-axis and the line $l$, shown shaded in Figure 5, is rotated through $360°$ about the $x$-axis.

Given that the volume of the solid formed is $50\pi$ units$^3$

find the value of $b$

(7)

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## 3. Question 11 · part whole · 7 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

**Question 11 continued**

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## 3. Question 11 · part whole · 7 marks

2025 JUN · Paper 01

Further practice · Attempt part whole only.

**Question 11 continued**

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## 3. Question 11 · part whole · 7 marks · Working and answer

Working and answer

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## 4. Question 2 · part whole · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

**2** The region enclosed by the curve with equation $y = e^{3x}$, the $x$-axis, the $y$-axis and the line with equation $x = 3$ is rotated through 360° about the $x$-axis.

Use algebraic integration to find, in terms of $\pi$ and e, the volume of the solid generated.

**(4)**

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## 4. Question 2 · part whole · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

**Question 2 continued**

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**(Total for Question 2 is 4 marks)**

## 4. Question 2 · part whole · 4 marks · Working and answer

Working and answer

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## 5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**7**

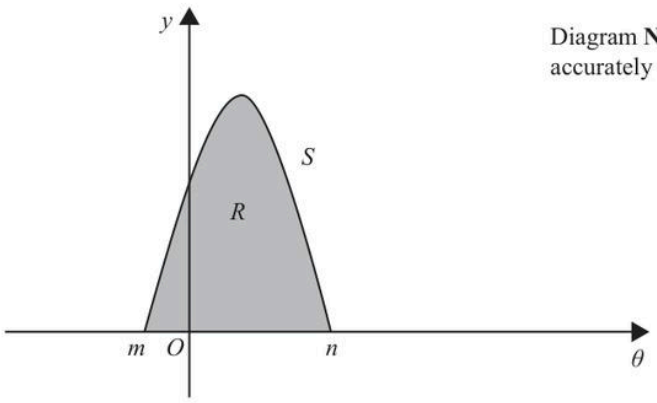



Diagram **NOT** accurately drawn

**Figure 2**

Figure 2 shows part of the curve $S$ with equation $y = \left(\cos 3\theta + \sqrt{3}\sin 3\theta\right)^{\frac{1}{2}}$

where $m \leqslant \theta \leqslant n$

The curve $S$ meets the $x$-axis at the point with coordinates $(m, 0)$ and at the point with coordinates $(n, 0)$

(a) Find the exact value of $m$ and the exact value of $n$

(3)

The finite region $R$, shown shaded in Figure 2, is bounded by the curve $S$, and the $x$-axis in the region $m \leqslant \theta \leqslant n$

The region $R$ is rotated through $2\pi$ radians about the theta-axis.

(b) Use calculus to find the exact volume of the solid generated.

(4)

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## 5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**Question 7 continued**

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## 5. Question 7 · part b · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

**Question 7 continued**

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Further practice · Attempt part b only.

**Question 7 continued**

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**(Total for Question 7 is 7 marks)**

## 5. Question 7 · part b · 4 marks · Working and answer

Working and answer

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## 6. Question 11 · part b · 6 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

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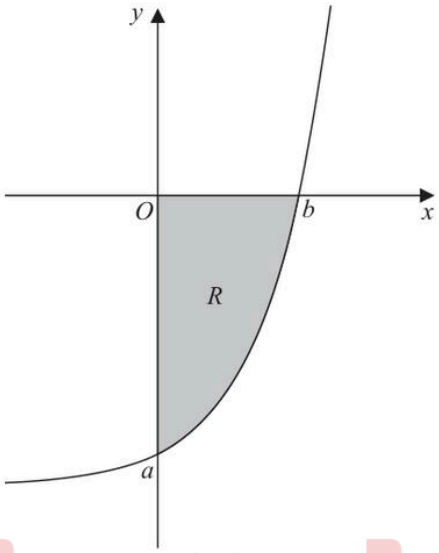



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**Figure 3**

The finite region $R$, shown shaded in Figure 3, is bounded by the curve with equation $y = e^{2x} - 9$ and the coordinate axes.

The curve crosses the coordinate axes at the points with coordinates $(0, a)$ and $(b, 0)$

(a) (i) Find the value of $a$

(ii) Show that $b = \ln 3$

**(3)**

The region $R$ is rotated through 360° about the $x$-axis.

(b) Use calculus to find the volume of the solid generated.

Give your answer in the form $\pi(p \ln 3 + q)$, where $p$ and $q$ are integers.

**(6)**

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## 6. Question 11 · part b · 6 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

**Question 11 continued**

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**Question 11 continued**

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## 6. Question 11 · part b · 6 marks · Working and answer

Working and answer

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